Laboratory Studies

Unless the diagnosis is certain, important investigations for all photosensitive conditions include assessment of circulating anti-nuclear factor and extractable nuclear antibody titers, as well as blood, urine, and stool porphyrin concentrations. Lesional histologic features are characteristic in many photodermatoses—particularly polymorphic light eruption (PMLE), hydroa vacciniforme (HV), and chronic actinic dermatitis (CAD)—but are rarely diagnostic except in HV. Direct immunofluorescence may assist in diagnosing cutaneous lupus.

Phototesting of normal back skin using a monochromator in CAD and solar urticaria (SU) often reproduces the papules or wheals typical of the condition, frequently at low irradiation doses, and helps define the action spectrum of the eruption. In xeroderma pigmentosum (XP), phototesting typically shows delayed erythema developing over 2–3 days with an abnormally low dose threshold, often progressing to blistering. However, phototesting is only moderately reliable in other photodermatoses, although solar-simulating or broadband irradiation may help induce the rash to support diagnosis.

In eczematous photosensitivity, patch and photopatch testing are essential to identify potential inducing or exacerbating allergens. For certain genophotodermatoses, specialized techniques such as assessment of DNA excision repair or RNA synthesis recovery rate in cultured patient fibroblasts after ultraviolet radiation (UVR) exposure are required for definitive diagnosis.

Phototesting

The use of phototesting in clinical dermatology varies widely between countries and centers. While it is the investigation of choice when the diagnosis of a photodermatosis is uncertain or when details of the action spectrum are needed, it remains primarily a research tool available at only a limited number of clinical centers.

Phototesting in suspected photodermatosis falls into two main categories: - **Monochromatic phototesting**: Performed on the upper back at a series of wavelengths and doses to elicit the eruption and identify the action spectrum. - **Photoprovocation testing**: Uses a broad-spectrum source to induce the eruption for clinical diagnosis or biopsy when necessary.

Monochromatic phototesting is best conducted with a xenon arc irradiation monochromator for precise determination of wavelength dependency. Photoprovocation testing is most effectively performed with a solar simulator—a filtered xenon arc source that closely mimics natural sunlight at noon during midsummer in temperate regions. Alternative protocols using filtered metal halide or fluorescent light sources are also acceptable. Sunlight with filters has been used in some settings, but its variability makes it unsuitable for routine clinical use.

Monochromatic testing should be performed on unaffected skin of the upper back, lateral to the paravertebral groove. Lesion induction—except in conditions like SU and CAD, where monochromatic testing readily provokes lesions—is best achieved with broadband sources over larger areas of skin known to be susceptible. Conditions such as PMLE, actinic prurigo (AP), and HV often require repeated irradiation with UVA, UVB, or combined sources to reproduce the characteristic skin changes.

Precautions and Standardization

To avoid false-negative results, potent topical and systemic corticosteroids should be discontinued for several days prior to testing if possible. The effect of oral immunosuppressive agents on test outcomes is less clear, but these should also ideally be withheld. False-positive results may occur in patients with widespread active disease; therefore, the eruption should be well-controlled before testing, ideally by maintaining the patient in a low-light environment.

However, these optimal conditions are often difficult to achieve during active eruptions, and testing may proceed with the understanding that results could be unreliable.

All phototesting must be performed using standardized, sequentially increasing doses (often in a geometric series) and wavelengths. Results should be read at consistent time points post-exposure under controlled environmental conditions of light and temperature to ensure accuracy and reproducibility.